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Findings From the 2021 Survey of Health and Recreation in Washington State: *Gambling Behaviors and Prevalence*

In 2020, the Washington State Legislature directed the Washington State Health Care Authority (HCA) to conduct a survey of the prevalence of problem gambling among adult residents of the state.[1]

For this analysis, HCA defined problem gamblers as gamblers who score above a certain rate on a severity index. HCA contracted with the Washington State Institute for Public Policy (WSIPP) to analyze the results of the survey to provide information on the following:

- The prevalence of gambling and problem gambling among adults residents in Washington State;
- The distribution of gambling methods (online vs. brick and mortar);
- The prevalence of problem gambling with co-occurring disorders;
- Beliefs and attitudes towards gambling;
- Methods of outreach and awareness reaching different demographic groups; and
- Beliefs and attitudes about gambling.

This report is presented in three sections. Section I provides the background and methodology of the survey. Section II presents findings from WSIPP's analysis for each of the study questions. Section III provides a summary and examines the limitations of the report.

Summary

The 2021 survey of adults in Washington State found the following:

- 44.7% reported they had gambled in the past 12 months.
- Of those who had gambled, 3.5% were classified as problem gamblers (representing 1.4% of all adult residents of Washington State).
- Statistically significant differences were detected for the prevalence of gambling and problem gambling among some different demographic populations (demographics collected include gender, marital status, ethnicity, age, education, military service, employment, type of insurance, and geographic region).
- Of those adults who gambled, 27.8% gambled online. Compared to those who gambled only in brick-and-mortar establishments, online gamblers are significantly more likely to be problem gamblers (7.8% vs 1.7%).
- Gamblers who self-identified as having problems with substance use, mental health, or other behaviors, were more likely to be problem gamblers than others not reporting these problems.
- Most of the population (67.8%) said they thought the harms of gambling outweighed the benefits. A similar proportion (64.2%) said the availability in Washington was fine—neither too available nor not available enough.

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[1] Engrossed Substitute Senate Bill 6168, Chapter 357, Laws of 2020. HCA used WSIPP's report in the creation of their report to the legislature.

I. Background & Survey Methods

The Health Care Authority (HCA) contracted with the Social and Economic Sciences Research Center (SESRC) at Washington State University to conduct the survey. To obtain a representative sample of adults in Washington, the survey used a random sample of postal addresses. The sample was based on the U.S. Postal Service Delivery Sequence File.

Respondents living at the randomly selected addresses were contacted by mail to complete a web survey. The adult living at the address who had the most recent birthday was asked to complete the survey. A cash $1 pre-incentive was included in the first contact. Non-respondents were offered a paper questionnaire as an alternative mode for completing the survey.

Of the 52,000 randomly selected residential addresses, 2,875 were determined to be ineligible (either no one was living at the address, no one was a Washington resident within the past year, etc.). For this study, 9,249 of the 49,125 eligible respondents completed or partially completed the survey including eight questions that screened for gambling behavior in the past year.[2] The final response rate was 18.8%.

To boost response from the potential Spanish-speaking population, SESRC, in consultation with HCA, sent a Spanish language paper questionnaire to the addresses in the sample that were flagged as potentially Spanish speaking.

For the survey to represent the adult population of Washington, final responses were weighted on gender, race, Hispanic ethnicity, education, age, and marital status. When the information was missing, values were imputed so that the weighted sample represented the state population. In some cases, it was necessary to combine groups. For example, SESRC combined race responses to create just two categories: "White" and "non-White."[3] This allowed for sufficient sample sizes for analysis.

To investigate possible regional differences in gambling behavior we provide an analysis of the geographical regions where state-paid behavioral health services are provided. In Washington, there are ten regional Managed Care Organizations (MCO, also referred to as "integrated managed care"). To allow for more accurate estimates for these geographical regions, separate weights were also created to correctly weight within each MCO.[4] A map showing these regions is provided in Exhibit 1.

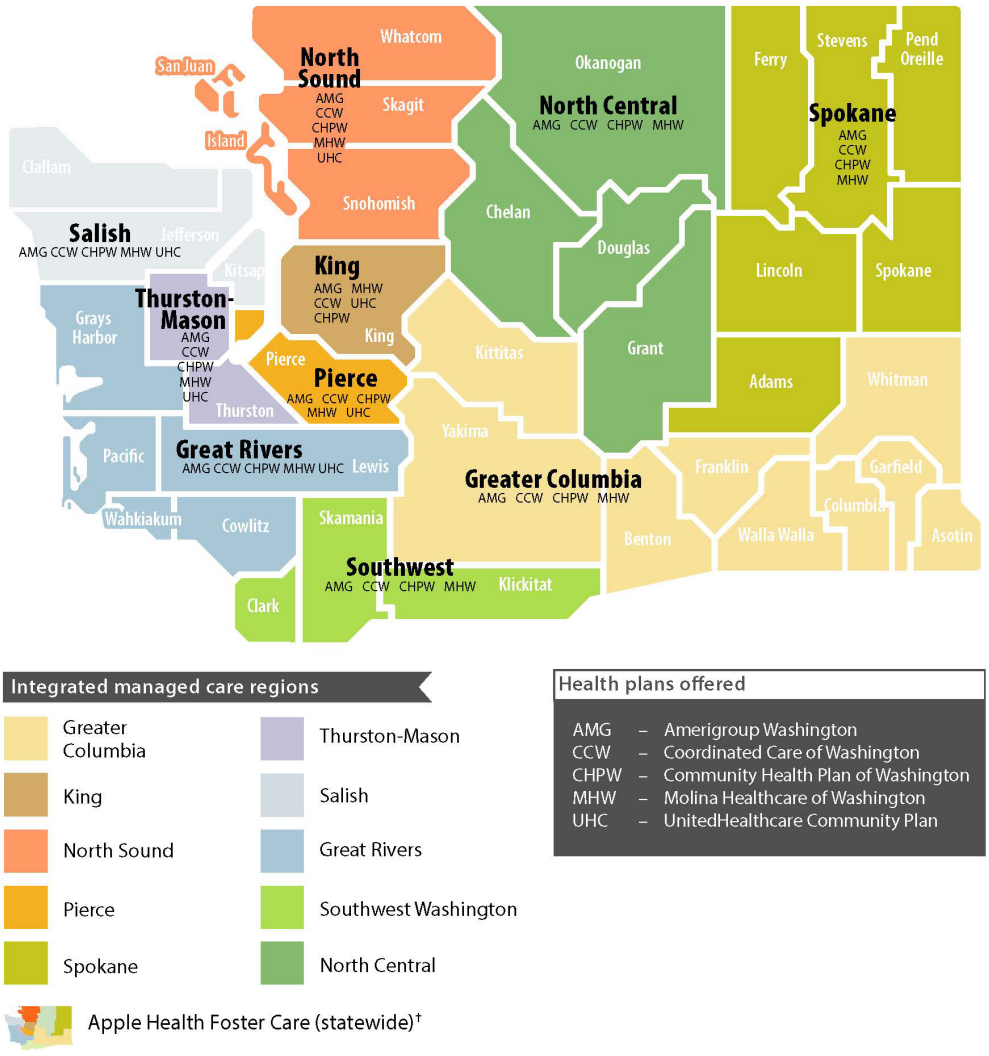
[2] The questions asked about the following activities in the past 12 months: 1) purchased lottery tickets; 2) gambled at a card room; 3) gambled at a tribal casino; 4) gambled on horseracing; 5) gambled online; 6) gambled using pull-tabs, bingo, or raffles; 7) done any other types of gambling or games of chance; and 8) traveled out of state to gamble.

3 Throughout the rest of this report, we report race as "White" and "people of color (POC)." The designation "POC" includes those participants who selected "Black or African American," "Asian," "Native Hawaiian or Other Pacific Islander," "American Indian/Alaska Native/Tribal," or "Other." Hispanic ethnicity is analyzed separately from race.

[4] More information about how the survey was conducted, the questionnaire, weighting, and responses to all questions is available at the Health Care Authority website.

Exhibit 1

Areas of Managed Care Organizations



† Apple Health Foster Care is a statewide program. Integrated managed care is provided through Apple Health Core Connections (Coordinated Care of Washington - CCW).

Notes:
Apple Health managed care service area map – January 2022.
Source: Washington State Health Care Authority.

If respondents indicated that they had done any type of gambling in the past 12 months, they were flagged as "gamblers."

Gamblers were directed to answer the nine questions of the Problem Gambling Severity Index (PGSI) (see Exhibit 2). Responses to each question are scored from 0 to 3. The total from all questions is used to identify problem gamblers. For this analysis, HCA defined "problem gamblers" as those scoring a 5 or greater on the PGSI.

Exhibit 2

Problem Gambling Severity Index (PGSI)

Please mark a response for each row	**Never**	**Sometimes**	**Most of the time**	**Almost always**
Have you bet more than you could really afford to lose?	0	1	2	3
Have you felt guilty about the way you gamble or what happens when you gamble?	0	1	2	3
Have you needed to gamble with larger amounts of money to get the same feeling of excitement?	0	1	2	3
Did you go back another day to try to win back the money you lost?	0	1	2	3
Have you borrowed money or sold anything to get money to gamble?	0	1	2	3
Has your gambling caused any financial problems for you or your household?	0	1	2	3
Has your gambling caused you any health problems, including stress or anxiety?	0	1	2	3
Have people criticized your betting or told you that you had a gambling problem, regardless of whether or not you thought it was true?	0	1	2	3
Have you felt that you might have a problem with gambling?	0	1	2	3

II. WSIPP Analysis

SESRC provided WSIPP with the final survey dataset. WSIPP analyzed the data using special survey protocols in SAS and Stata to appropriately estimate weighted population prevalence in all our analyses. Thus, although the group of adults surveyed was small (N = 9,249) relative to Washington's entire population, the survey was conducted and analyzed in a way that our findings represent Washington as a whole.

This section addresses five study questions that WSIPP was asked to address. We were unable to address several additional questions related to treatment for problem gambling as the prevalence rate was too low. That is, there were too few individuals who met the criteria for problem gamblers to observe meaningful differences in any treatment they sought.

Study Question 1—What is the prevalence of problem gambling among adults in WA State?

The statewide prevalence of gambling and problem gambling is displayed in Exhibit 3. At the time of this study (July through September 2021), we found that 44.7% of the adults in the state had engaged in gambling in the past year. This is a lower prevalence rate than was observed in two previous studies in Washington. A survey conducted in 1998 found that 74% of adults had gambled in the past year.[5] The second study done in 2003 found that 54% of adults had gambled in the past year.[6]

While methods used to assess gambling behaviors were not identical across the three surveys, we e attribute this decline in gambling prevalence between 2003 and 2021, at least in part, to the COVID-19 pandemic when brick and mortar gambling venues were closed for many months.

Based on this survey, we estimate that 1.4% of the adult population in Washington are problem gamblers, comparable to rates observed in the two earlier Washington studies. Among those who gamble, we estimate that 3.5% are problem gamblers.

Who Gambles?

Exhibit 3 provides the prevalence rates for several demographics. The table provides information on the percentage of each population (e.g., men and women) who gamble and are problem gamblers. It also indicates rates of problem gambling, limiting the analysis to only those who gamble.

Regional analysis was completed using Apple Health regional service areas, as shown in Exhibit 1.

[5] Volberg, R.A., & Moore, W.E. (1998). *Gambling and Problem Gambling In Washington State: A Replication Study, 1992 to 1998.* Report to the Washington State Lottery.

[6] Mancuso, D., Gilson, M., & Felber, B. (2005). *The 2003 Washington State Needs Assessment Household Survey: Substance Use, Substance Use Disorders, and Need for Treatment in Washington State.* Chapter 6. Problem Gambling, Substance Use, and Treatment Need. Olympia WA, Department of Social and Health Services.

We found that prevalence rates for gambling varied significantly by the following:

- Gender,
- Marital status,
- Hispanic ethnicity,
- Age,
- Level of education,
- Military service,
- Employment,
- Health insurance, and
- Geographic region.

Problem gambling varied significantly by marital status, education, and employment status.

While women were less likely to gamble than men, among gamblers, the rates of problem gambling did not differ by gender.

Prevalence of gambling and problem gambling did not vary by physical disability status.

How to Read Exhibits 3-13

In this report, each row in Exhibits 3-13 represents a population. For example, in Exhibit 3, the full sample represents 100% of the population. Of the population, 44.7% gambled in the past 12 months. Of the entire population, we estimate 1.4% are problem gamblers. Among just the gamblers, 3.5% are problem gamblers.

The tables provide the point estimate and the margin of error. If the survey were repeated many times, we would expect the point estimate to lie within the bounds of the margin of error 95% of the time.

When ranges around point estimates for different groups overlap, they may or may not be statistically different from one another. Thus, we tested for significant differences between groups. For each demographic, we chose a "reference" population; findings for other populations were compared to the reference group, noted in the first column as "(ref)," by running a z-test of proportions. This test is mathematically equivalent to a Chi-square test of homogeneity. Running either of these identical tests is the standard for comparing group means across categorical variables.

Where differences between populations are statistically significantly different ($p \leq 0.05$), we indicate the significance with **bolded text.** For example, in Exhibit 3, we observe that those who were never married are significantly less likely to be gamblers (39.6%) than those who were married (46.2%), whereas those with an "other" marital status were no more or less likely (47.3%) than those who were married.

Because the prevalence rate for problem gambling is very low, we sometimes see that the number of problem gamblers in a demographic is too small (fewer than 10 respondents) to estimate prevalence reliably. We indicate this as "N too small."

In some instances, each demographic category is different from each and every one of the others. In that case, we indicate this with a "ŧ" symbol and *italics*.

Weighted Ns in tables may vary because not all respondents answered all questions.

Lastly, we only analyzed the demographic subsets requested by HCA.

Exhibit 3

Prevalence of Gambling and Problem Gambling

Population	Weighted N	Full sample (N=9,249)			Gamblers only (N=4,130)
		Percent of sample	Percent gamblers	Percent problem gamblers	Percent problem gamblers
Full sample	9,249	100	44.7 (± 1.5)	1.4 (± 0.4)	3.5 (± 1.0)
Gender [#]					
Men (ref)	4,576	49.5 (± 1.5)	46.2 (± 2.3)	1.3 (± 0.6)	3.0 (± 1.3)
Women	4,673	50.5 (± 1.5)	**43.1 (± 2.0)**	1.5 (± 0.6)	4.1 (± 1.5)
Race					
White (ref)	7,195	77.8 (± 1.4)	44.9 (± 1.7)	1.2 (± 0.4)	2.8 (± 0.9)
POC	2,054	22.2 (± 1.4)	43.9 (± 3.6)	2.3 (± 1.2)	**6.0 (± 3.0)**
Marital status					
Married (ref)	4,881	52.8 (± 1.5)	46.2 (± 1.9)	1.0 (± 0.4)	2.5 (± 1.0)
Never married	2,516	27.2 (± 1.6)	**39.6 (± 3.7)**	1.8 (± 1.1)	4.8 (± 2.7)
Other	1,852	20.0 (± 1.0)	47.3 (± 2.7)	**2.0 (± 0.9)**	4.7 (± 2.0)
Ethnicity					
Hispanic (ref)	918	9.9 (± 1.2)	51.5 (± 6.3)	2.7 (± 2.2)	5.7 (± 4.5)
Non-Hispanic	8,331	90.1 (± 1.2)	**43.9 (± 1.5)**	1.3 (± 0.4)	3.3 (± 0.9)
Age group					
18-34 years (ref)	2,794	30.2 (± 1.6)	41.0 (± 3.5)	2.1 (± 1.1)	5.4 (± 2.8)
35-64 years	4,640	50.2 (± 1.5)	**48.1 (± 2.0)**	1.1 (± 0.4)	2.6 (± 1.0)
65+ years	1,815	19.6 (± 0.9)	41.5 (± 2.3)	1.1 (± 0.5)	3.3 (± 1.5)
Education					
High school or less	2,926	31.6 (± 1.7)	**46.1 (± 3.7)**	1.4 (± 0.9)	3.8 (± 2.2)
Some college	3,214	34.8 (± 1.4)	**48.8 (± 2.4)**	**2.1 (± 0.8)**	**4.7 (± 1.7)**
BA degree (ref)	1,975	21.4 (± 0.9)	41.6 (± 2.1)	0.9 (± 0.4)	2.3 (± 1.1)
Advanced degree	1,134	12.3 (± 0.6)	**34.5 (± 2.2)**	**0.3 (± 0.3)**	N too small
Physical disability*					
Yes (ref)	681	6.1 (± 0.7)	43.8 (± 5.5)	2.3 (± 1.9)	6.0 (± 4.6)
No	8,443	93.9 (± 0.7)	44.5 (± 1.6)	1.3 (± 0.4)	3.3 (± 1.0)
Military service					
Any military service (ref)	943	10.5 (± 0.9)	51.5 (± 4.3)	1.2 (± 0.7)	2.7 (± 1.6)
No military service	8,065	89.5 (± 0.9)	**43.5 (± 1.6)**	1.3 (± 0.4)	3.2 (± 1.0)
Employment					
Employed (ref)	5,062	57.2 (± 1.5)	47.4 (± 2.1)	1.2 (± 0.5)	2.9 (± 1.1)
Unemployed/other	1,960	22.2 (± 1.5)	**38.8 (± 3.9)**	2.2 (± 1.1)	6.0 (± 3.0)
Retired	1,827	20.6 (± 1.0)	**42.4 (± 2.4)**	**0.6 (± 0.4)**	1.6 (± 1.0)

Exhibit 3 (cont.)

Prevalence of Gambling and Problem Gambling

Population	Weighted N	Full sample (N=9,249)		Gamblers only (N=4,130)	
		Percent of sample	Percent gamblers	Percent problem gamblers	Percent problem gamblers
Full sample	9,249	100	44.7 (± 1.5)	1.4 (± 0.4)	3.5 (± 1.0)
Type of health Insurance					
Private insurance (ref)	5,132	59.7 (± 1.6)	46.3 (± 2.0)	1.0 (± 0.4)	2.4 (± 1.0)
Medicare/Medicaid	2,590	30.1 (± 1.5)	**40.8 (± 2.9)**	1.6 (± 0.8)	4.3 (± 2.1)
Uninsured/other	869	10.1 (± 1.1)	**41.9 (± 5.9)**	2.3 (± 1.9)	5.9 (± 4.7)
MCO region					
King (ref)	2,860	30.5 (± 1.4)	38.7 (± 2.5)	1.2 (± 0.7)	3.5 (± 2.0)
Great Rivers	329	3.9 (± 0.6)	**48.0 (± 8.4)**	2.9 (± 2.7)	N too small
Greater Columbia	794	9.2 (± 0.9)	**48.7 (± 5.4)**	0.8 (± 0.9)	N too small
North Central	275	3.2 (± 0.6)	44.2 (± 9.7)	0.5 (± 1.0)	N too small
North Sound	1,644	17.0 (± 1.1)	**45.1 (± 3.6)**	1.3 (± 0.8)	3.3 (± 1.8)
Pierce	915	11.6 (± 1.0)	**49.7 (± 4.8)**	1.7 (± 1.1)	3.9 (± 2.6)
Salish	545	5.2 (± 0.6)	44.9 (± 6.1)	2.2 (± 2.5)	N too small
Southwest	613	6.7 (± 0.8)	**51.1 (± 5.7)**	0.7 (± 0.8)	N too small
Spokane	783	8.1 (± 0.8)	**46.0 (± 5.1)**	2.0 (± 1.9)	N too small
Thurston-Mason	491	4.7 (± 0.6)	45.6 (± 6.7)	1.2 (± 1.7)	N too small

Notes:

For demographics, individual rows are calculated as a percentage of the total for the row only, not as a percentage of the column header.

[#] Gender was reported as "male" or "female." Throughout this report, we report as "men" and "women."

* Physical disability was determined from the response to the question, "Do you now have any health problem that requires you to use special equipment, such as a cane, a wheelchair, a special bed, or a special telephone?"

Bold = Significance at the 0.05-level.

POC = People of color.

MCO = Managed Care Organization.

Study Question 2—What is the distribution of gambling methods (e.g., online vs. brick and mortar) that are being utilized by WA State adults?

The survey inquired about gambling venues and online methods of gambling. It asked a generic question about gambling online. Respondents who indicated they did gamble online, in addition to those who endorsed any of the following, were categorized as "online gamblers."[7]

- Social casinos online,
- Sports betting online,
- Gambling in cryptocurrency,
- Purchasing chances to win in video games,
- Gambling on mobile phones/tablets, and
- Purchasing a chance to "uplevel" in online games.

We categorized respondents to be brick-and-mortar gamblers if they endorsed any of the following:

- Game rooms,
- Casinos,
- Horse track/parimutuel betting,
- Purchasing lottery tickets,
- Traveling out of state to gamble, and
- Other types of gambling or games of chance.

We observed that most online gamblers also gambled in brick-and-mortar establishments. For this analysis, we split gamblers into those who did *any* online gambling ("online gamblers") and those who gambled only at brick-and-mortar establishments ("brick-and-mortar gamblers").

As can be seen in Exhibit 4, those who gamble online are significantly more likely to be problem gamblers than those who only gamble in brick-and-mortar establishments.

In Exhibits 5 and 6, we provide the differences in demographics of online and brick-and-mortar gambling.

Exhibit 4

Comparison of Online and Brick-and-Mortar Gamblers

	Number of gamblers	Percent (±) margin of error	
		Percent gamblers	Percent problem gamblers
Online	1,147	27.8 (± 2)	**7.8 (± 2.9)**
Brick and mortar	2,982	72.2 (± 2)	1.7 (± 0.7)

Note:
Bold = Significance at the 0.05-level.

[7] About 3% of the sample did not endorse any of the gambling screening questions and, thus, were not classified as gamblers but later in the survey they indicated that they did some sort of online gambling. Because they did not complete the PGSI questions we are unable to speak to the severity of gambling in this group, they were omitted from our analysis.

Online Gamblers

In our analysis, we find that the prevalence of online gambling is higher among men than women (14.0% vs 10.8%).

Those aged 18 to 34 were most likely to gamble online, followed by those aged 35 to 64. Those over age 65 were least likely to be online gamblers. Individuals who were never married were more likely than those who were currently married to gamble online, and those with an advanced degree were less likely than those with a high school diploma to gamble online. We see similar patterns when the sample is restricted to gamblers only. We also observed differences in online gambling by military service, employment status, and geographic region.

Prevalence of problem gambling among online gamblers differed by marital status, age group, and employment status.

Exhibit 5

Online Gamblers

	Number of online gamblers	Percent (±) margin of error		
		Percent of full sample (N=9,249)	Online as percent of all gamblers (N=4,130)	Problem gamblers as percent online gamblers (N=1,147)
Full sample	1,147	12.4 (± 1.1)	27.8 (± 2.2)	7.8 (± 2.9)
Gender				
Men (ref)	642	14.0 (± 1.7)	30.4 (± 3.3)	6.9 (± 3.9)
Women	505	**10.8 (± 1.3)**	**25.1 (± 2.8)**	9.0 (± 4.2)
Race				
White (ref)	857	11.9 (± 1.2)	26.5 (± 2.3)	7.1 (± 3.0)
POC	290	14.1 (± 2.7)	32.2 (± 5.3)	10.0 (± 7.0)
Marital status				
Married (ref)	528	10.8 (± 1.2)	23.4 (± 2.5)	3.7 (± 2.3)
Never married	372	**14.8 (± 2.8)**	**37.4 (± 5.9)**	**11.4 (± 7.1)**
Other	247	13.3 (± 2.2)	**28.2 (± 4.1)**	**11.6 (± 6.3)**
Ethnicity				
Hispanic	146	15.9 (± 4.7)	30.9 (± 8.4)	N too small
Non-Hispanic (ref)	1,001	12.0 (± 1.1)	27.4 (± 2.2)	7.4 (± 2.9)
Age group				
18 to 34 years (ref)	465	*16.7 (± 2.7)†*	*40.6 (± 5.5)†*	10.9 (± 6.0)
35 to 65 years	567	*12.2 (± 1.4)†*	*25.4 (± 2.6)†*	6.1 (± 3.1)
65+ years	115	*6.3 (± 1.1)†*	*15.2 (± 2.6)†*	**3.6 (± 3.5)**
Education				
High school or less	395	13.5 (± 2.6)	29.3 (± 5.1)	N too small
Some college	437	13.6 (± 1.8)	27.9 (± 3.2)	11.6 (± 5.4)
BA degree (ref)	233	11.8 (± 1.5)	28.3 (± 3.1)	N too small
Advanced degree	82	**7.2 (± 1.3)**	**21.0 (± 3.4)**	N too small
Physical disability				
Yes (ref)	64	11.4 (± 3.5)	26.1 (± 7.3)	N too small
No	1,075	12.5 (± 1.1)	28.2 (± 2.3)	7.0 (± 2.9)
Military service				
Any military service (ref)	107	11.4 (± 3.0)	22.1 (± 5.4)	N too small
No military service	1,025	12.7 (± 1.2)	**29.2 (± 2.4)**	7.7 (± 3.1)
Employment				
Employed (ref)	760	15.0 (± 1.6)	31.7 (± 3.0)	5.7 (± 3.0)
Unemployed/other	246	12.5 (± 2.6)	32.3 (± 6.0)	**15.8 (± 8.9)**
Retired	103	**5.6 (± 1.1)**	**13.3 (± 2.5)**	N too small

Exhibit 5 (cont.)
Online Gamblers

	Number of online gamblers	Percent (±) margin of error		
		Percent of full sample (N=9,249)	Online as percent of all gamblers (N=4,130)	Problem gamblers as percent online gamblers (N=1,147)
Full sample	1,147	12.4 (± 1.1)	27.8 (± 2.2)	7.8 (± 2.9)
Type of health insurance				
Private insurance (ref)	671	13.1 (± 1.4)	28.3 (± 2.7)	5.2 (± 2.8)
Medicare/Medicaid	276	10.7 (± 2.1)	26.1 (± 4.6)	9.9 (±7.1)
Uninsured/other	130	14.9 (± 4.5)	35.6 (± 9.2)	N too small
MCO region				
King (ref)	333	11.6 (± 1.8)	30.0 (± 4.0)	8.0 (± 5.6)
Great Rivers	40	12.3 (± 5.4)	25.5 (± 10.4)	N too small
Greater Columbia	103	**26.6 (± 7.5)**	26.6 (± 7.5)	N too small
North Central	37	13.3 (± 6.5)	30.1 (± 12.7)	N too small
North Sound	198	12.0 (± 2.4)	26.7 (± 4.9)	6.7 (± 5.4)
Pierce	125	13.7 (± 3.4)	27.5 (± 6.2)	10.3 (± 8.2)
Salish	76	13.9 (± 5.4)	31.0 (± 10.3)	N too small
Southwest	81	13.2 (± 4.0)	25.7 (± 7.2)	N too small
Spokane	77	9.8 (± 3.3)	**21.3 (± 6.6)**	N too small
Thurston-Mason	66	13.4 (± 5.5)	29.4 (± 10.5)	N too small

Notes:
For demographics, individual rows are calculated as a percentage of the total for the row only, not as a percentage of the column header
† *indicates that all demographic groups in a column are significantly different from each other.*
Bold = Significance at the 0.05-level.
POC = People of color.
MCO = Managed Care Organization.

Exhibit 6 displays the findings from the analysis of those who gamble only at brick-and-mortar establishments. We find the percentage varies by gender, marital status, age group, education, military service, employment, type of health insurance, and geographic region, particularly when we restrict the sample to individuals who gamble at all

For example, those aged 35 and older were significantly more likely than individuals in the youngest age group to gamble in brick-and-mortar establishments. Those who are unemployed are least likely to be brick-and-mortar gamblers, followed by employed adults; retired people are the most likely. Finally, those with any military service history are more likely to gamble in brick-and-mortar establishments than those with no military service history.

We observed no demographic differences in the rate of problem gambling among brick-and mortar gamblers.

Exhibit 6

Brick-and-Mortar Gamblers

	Number of brick-and-mortar gamblers	Percent (±) margin of error		
		Percent of full sample (N=9,249)	Brick-and-mortar as percent all gamblers (N=4,130)	Problem gamblers as percent brick-and-mortar (N=1,147)
Full sample	2,982	32.2 (± 1.4)	72.2 (± 2.2)	1.7 (± 0.7)
Gender				
Men (ref)	1,473	32.2 (± 2.1)	69.6 (± 3.3)	1.2 (± 0.7)
Women	1,509	32.3 (± 1.9)	**74.9 (± 2.8)**	2.2 (± 1.2)
Race				
White (ref)	2,371	33.0 (± 1.5)	73.5 (± 2.3)	1.3 (± 0.5)
POC	902	29.8 (± 3.2)	67.8 (± 5.3)	3.6 (± 2.7)
Marital status				
Married (ref)	1,729	35.4 (± 1.8)	*76.6 (± 2.5)†*	1.1 (± 1.0)
Never married	624	**24.8 (± 3.2)**	*62.6 (± 5.9)†*	1.9 (± 1.1)
Other	629	34.0 (± 2.6)	*71.8 (± 4.1)†*	1.9 (± 1.1)
Ethnicity				
Hispanic (ref)	326	35.6 (± 6.1)	69.1 (± 8.4)	N too small
Non-Hispanic	2,656	31.9 (± 1.4)	72.6 (± 2.2)	1.5 (± 0.5)
Age group				
18 to 34 years (ref)	680	24.3 (± 3.0)	*59.4 (± 5.5)†*	N too small
35 to 65 years	1,664	**35.9 (± 1.9)**	*74.6 (± 2.6)†*	1.3 (± 0.6)
65+ years	639	**35.2 (± 2.3)**	*84.8 (± 2.6)†*	3.2 (± 1.7)
Education				
High school or less	953	32.6 (± 3.4)	70.7 (± 5.1)	N too small
Some college	1,131	35.2 (± 2.2)	72.1 (± 3.2)	2.0 (± 0.9)
BA degree (ref)	589	29.8 (± 1.9)	71.7 (± 3.1)	1.8 (± 1.2)
Advanced degree	310	27.3 (± 2.1)	**79.0 (± 3.4)**	N too small
Physical disability				
Yes (ref)	182	32.4 (± 5.1)	73.9 (± 7.3)	N too small
No	2,744	32.0 (± 1.5)	71.8 (± 2.3)	1.8 (± 0.7)
Military service				
Any military service (ref)	378	40.1 (± 4.2)	77.9 (± 5.4)	N too small
No military service	2,483	**30.8 (± 1.5)**	**70.8 (± 2.4)**	1.2 (± 0.5)
Employment				
Employed (ref)	1,637	*32.3 (± 1.9)†*	68.3 (± 3.0)	1.3 (± 0.6)
Unemployed/other	514	*26.2 (± 3.5)†*	67.7 (± 6.0)	N too small
Retired	671	*36.7 (± 2.4)†*	**86.7 (± 2.5)**	N too small

Exhibit 6 (cont.)

Brick-and-Mortar Gamblers

	Number of brick-and-mortar gamblers	Percent (±) margin of error		
		Percent of full sample (N=9,249)	Brick-and-mortar as percent all gamblers (N=4,130)	Problem gamblers as percent brick-and-mortar (N=1,147)
Full sample	2,982	32.2 (± 1.4)	72.2 (± 2.2)	1.7 (± 0.7)
Type of health insurance				
Private insurance (ref)	1,705	33.2 (± 1.9)	71.7 (± 2.7)	1.1 (± 0.6)
Medicare/Medicaid	781	30.2 (± 2.6)	73.9 (± 4.6)	2.4 (± 1.3)
Uninsured/other	234	**26.9 (± 5.2)**	64.4 (± 9.2)	N too small
MCO region				
King (ref)	775	27.1 (± 2.3)	70.0 (± 4.0)	1.5 (± 1.0)
Great Rivers	329	**35.7 (± 8.0)**	74.5 (± 10.4)	1.5 (± 2.3)
Greater Columbia	284	**35.8 (± 5.1)**	73.4 (± 7.5)	1.4 (± 1.5)
North Central	37	30.8 (± 8.6)	69.9 (± 12.7)	N too small
North Sound	543	**33.0 (± 3.3)**	73.3 (± 4.9)	1.9 (± 1.4)
Pierce	330	**36.1 (± 4.5)**	72.5 (± 6.2)	N too small
Salish	169	31.0 (± 5.4)	69.0 (± 10.3)	N too small
Southwest	233	**38.0 (± 5.5)**	74.3 (± 7.2)	N too small
Spokane	284	**36.2 (± 4.8)**	**78.7 (± 6.6)**	N too small
Thurston-Mason	158	32.2 (± 6.0)	70.6 (± 10.5)	N too small

Notes:

For demographics, individual rows are calculated as a percentage of the total for the row only, not as a percentage of the column header.

† indicates that all demographic groups in a column are significantly different from each other.

Bold = Significance at the 0.05-level.

POC = People of color.

MCO = Managed Care Organization.

Study Question 3—What is the prevalence of co-occurring disorders with problem gambling?

The survey was not designed to diagnose co-occurring disorders. To answer this question, we used responses to questions about problems with mental health, substance use, and other behavioral issues. We also examined the use of specific substances and physical health.

In Exhibit 7, which displays the findings, we list the weighted number of respondents in the survey sample, within each sub-sample, and within demographic groups. We then indicate the percentage within each demographic group, both overall and the percentage that falls within a given sub-sample. For example, 3.3% of the population endorsed having a problem with substance use. Of those with a substance use problem, 51% were gamblers and 8% were problem gamblers. Among gamblers with a substance use problem, 16.5% were identified as problem gamblers. This is significantly greater than we observed in the population without substance problems (3.0%).

We observed that individuals who self-identified as having a problem with substance use, mental health, or other behaviors were significantly more likely to be problem gamblers than others without the co-occurring problem.

Additionally, we found that individuals endorsing general substance use[8] were significantly more likely to gamble. Those reporting poor health, and those using tobacco, cannabis, and other drugs were significantly more likely to be problem gamblers.

Individuals reporting poor health were no more likely to gamble than those in good health. However, a higher percentage of those in poor health were problem gamblers.

We found that gambling status was not significantly different depending on a respondent's disability status.

[8] General substance use includes alcohol, tobacco, cannabis, and other drugs not intended for medical use.

Exhibit 7

Co-occurring Disorders with Problem Gambling

	Weighted N	Percent (±) margin of error Full sample (N=9,249)			Gamblers (N=4,130)
		Percent of sample	Percent gamblers	Percent problem gamblers	Percent problem gamblers
Full sample[a]	9,249	100	44.7 (± 1.5)	1.4 (± 0.4)	3.5 (± 1.0)
Drug or alcohol problem[b]					
No (ref)	8,847	96.7 (± 0.6)	44.3 (± 1.5)	1.2 (± 0.4)	3.0 (± 0.9)
Yes	306	3.3 (± 0.6)	51.0 (± 10.0)	**8.0 (± 6.2)**	**16.5 (± 11.9)**
Behavioral issues[c]					
No (ref)	7,992	87.6 (± 1.2)	43.9 (± 1.6)	0.8 (± 0.3)	2.1 (± 0.7)
Yes	1,137	12.5 (± 1.2)	48.8 (± 5.0)	**5.3 (± 2.5)**	**11.7 (± 5.2)**
Anxiety, depression, mental health problems[d]					
No (ref)	6,664	73.3 (± 1.5)	44.5 (± 1.7)	0.8 (± 0.3)	2.1 (± 0.7)
Yes	2,425	26.7 (± 1.5)	44.5 (± 3.3)	**3.1 (± 1.3)**	**7.5 (± 3.1)**
Alcohol use					
No (ref)	2,535	27.9 (± 1.5)	33.7 (± 3.0)	1.4 (± 0.8)	5.0 (± 2.7)
Yes	6,558	72.1 (± 1.5)	**48.8 (± 1.7)**	1.4 (± 0.5)	3.2 (± 1.1)
Tobacco use					
No (ref)	7,617	83.6 (± 1.3)	41.3 (± 1.6)	1.0 (± 0.4)	2.6 (± 0.9)
Yes	1,497	16.4 (± 1.3)	**60.3 (± 4.3)**	**3.5 (± 1.7)**	**6.4 (± 3.1)**
Cannabis use					
No (ref)	6,716	73.6 (± 1.4)	41.1 (± 1.7)	0.9 (± 0.3)	2.4 (± 0.9)
Yes	2,412	26.4 (± 1.4)	**53.8 (± 3.2)**	**2.9 (± 1.2)**	**5.9 (± 2.4)**
Any other drug use					
No (ref)	8,714	95.2 (± 0.7)	43.7 (± 1.5)	1.1 (± 0.4)	2.9 (± 0.9)
Yes	435	4.8 (± 0.7)	**58.7 (± 7.8)**	**7.4 (± 4.9)**	**13.1 (± 8.4)**
Physical health					
Good health (ref)	7,445	82.4 (± 1.3)	44.2 (± 1.7)	1.1 (± 0.4)	2.8 (± 0.9)
Poor health	1,588	17.6 (± 1.3)	46.4 (± 3.9)	**3.0 (± 1.6)**	**7.3 (± 3.8)**
Physical disability					
Yes (ref)	681	6.1 (± 0.7)	43.8 (± 5.5)	2.3 (± 1.9)	6.0 (± 4.6)
No	8,443	93.9 (± 0.7)	44.5 (± 1.6)	1.3 (± 0.4)	3.3 (± 1.0)

Notes:

[a] For demographics, individual rows are calculated as a percentage of the total for the row only, not as a percentage of the column header.

[b] Substance use stems from survey question 10: *Have you had any problems with drugs or alcohol in the past 12 months?*

[c] Behavioral issues stem from question 12: *Have you had any problems with other behaviors(s) in the past 12 months such as overeating, sex or pornography, exercise, internet chat lines, or other issues?*

[d] Anxiety, depression, and mental health stem from question 13: *In the past 12 months, have you had any serious problems with depression, anxiety, or other mental health problems?*

Bold = Significance at the 0.05-level.

Study Question 4—What methods of outreach and awareness are reaching different demographic groups?

Methods of Problem Gambling Outreach

The survey sought to uncover the efficacy of problem gambling outreach campaigns by asking respondents whether they had heard or seen information about problem gambling from any of the following sources:

- Radio,
- Television,
- Article,
- Brochure or poster at a gambling venue,
- From another person, or
- Other source of information.

As shown in Exhibit 8, television and radio were reported to be the most common sources where respondents heard information pertaining to problem gambling. Of respondents who heard about problem gambling, 19.4% heard from the television and 13.7% heard from the radio. In comparison, only 5% to 6% of respondents heard about problem gambling from an article, brochure, or another person, and only 3.6% heard from some other source.

Exhibit 8

Awareness of Problem Gambling Outreach

Source	Weighted N	Percent heard about problem gambling through source
Sample	9,219	100
Radio	1,259	**13.7 (± 1.0)**
Television	1,788	**19.4 (± 1.2)**
Article	578	6.3 (± 0.8)
Brochure (ref)	550	6.0 (± 0.7)
Another person	491	5.3 (± 0.8)
Other	332	**3.6 (± 0.5)**

Note:
Bold = Significance at the 0.05-level.

Next, we divided these six outreach categories into those classified as broadcast (radio and television) and those that were not (the rest).

Exhibits 9 and 10 display information about broadcast and other outreach, respectively. The tables provide information about respondents in the full survey sample and among those who gambled at least once within the last 12 months. The percentage of respondents within each demographic group and the proportion of those respondents who heard about problem gambling are given for both sample groups. For example, in Exhibit 9, 49.5% of the full sample are men. Of the men in the full sample, 30.8% heard about problem gambling through broadcast. Among men who gamble, 36% heard about problem gambling through broadcast.

Our results showed that among the full survey population, women, POC, Hispanics, and individuals 18 to 34 years old were significantly less likely to recall hearing about problem gambling through broadcast when compared to other groups in those demographic categories. When the sample was restricted to gamblers, race no longer significantly predicted one's likelihood of hearing about problem gambling through broadcast, but gender, ethnicity, and age did. Notably, gamblers were significantly more likely than non-gamblers to recall hearing about problem gambling through broadcast.

Exhibit 9

Broadcast Outreach

	Weighted N	Percent (±) margin of error			
		Full sample (N=9,219)		Gamblers (N=4,101)	
		Percent of population	Percent who heard a broadcast	Percent of population	Percent who heard a broadcast
Full sample	9,219	100	26.2 (± 1.3)	44.5 (± 1.5)	31.2 (± 1)
Gender					
Men (ref)	4,561	49.5 (± 1.5)	30.8 (± 2.1)	46.2 (± 3.0)	36.0 (± 3.2)
Women	4,658	50.5 (± 1.5)	**21.6 (± 1.6)**	43.1 (± 2.0)	**26.2 (± 2.6)**
Race					
White (ref)	7,169	77.8 (± 1.4)	27.4 (± 1.5)	44.9 (± 1.7)	32.4 (± 3)
POC	2,050	22.2 (± 1.4)	**22.0 (± 3.0)**	43.9 (± 3.6)	26.9 (± 4.8)
Ethnicity					
Hispanic (ref)	917	9.9 (± 1.2)	18.7 (± 4.6)	51.5 (± 6.3)	22.0 (± 7.1)
Non-Hispanic	8,302	90.1 (± 1.2)	**27.0 (± 1.4)**	**43.9 (± 1.5)**	**32.4 (± 2.1)**
Age group					
18-34 years (ref)	2,782	30.2 (± 1.6)	19.6 (± 2.8)	41.0 (± 3.4)	22.0 (± 4.5)
35-64 years	4,630	50.2 (± 1.5)	**29.3 (± 1.8)**	**48.1 (± 2.0)**	**35.1 (± 2.8)**
65+ years	1,807	19.6 (± 0.9)	**28.2 (± 1)**	41.5 (± 2.3)	**33.7 (± 3.6)**

Note:

For demographics, individual rows are calculated as a percentage of the total for the row only, not as a percentage of the column header.
Bold = Significance at the 0.05-level.
POC = People of color.

In Exhibit 10, we identified significant differences in the percentage of individuals who heard about problem gambling through other outreach methods across demographic groups. In the full sample, women, White respondents and individuals aged 35 and older were significantly less likely to hear about problem gambling through other outreach options. Among gamblers, women were less likely than men to have seen other outreach information. There were no other significant differences between demographic groups in the percentage of individuals who heard about problem gambling through outreach methods other than broadcast.

We observed no difference in the percentage of individuals who heard about problem gambling through other outreach methods between those of Hispanic and non-Hispanic ethnicities.

Exhibit 10

Other Outreach Methods

	Weighted N	Percent (range)			
		Full sample (N= 9,219)		Gamblers (N=4,101)	
		Percent of population	Percent who heard from other sources	Percent of population	Percent who heard from other sources
Full sample	9,219	100	17.3 (± 1.2)	44.5 (± 1.5)	20.0 (± 1.8)
Gender					
Men (ref)	4,561	49.5 (± 1.5)	19.5 (± 1.8)	51.2 (± 2.3)	22.2 (± 2.7)
Women	4,658	50.5 (± 1.5)	**15.1 (± 1.4)**	48.8 (± 2.3)	**17.7 (± 2.3)**
Race					
White (ref)	7,169	77.8 (± 1.4)	16.2 (± 1.2)	78.1 (± 2.0)	19.5 (± 1.9)
POC	2,050	22.2 (± 1.4)	**21.0 (± 3.0)**	21.9 (± 2.0)	21.8 (± 4.4)
Ethnicity					
Hispanic (ref)	917	9.9 (± 1.2)	18.3 (± 4.7)	11.5 (± 1.9)	19.1 (± 6.8)
Non-Hispanic	8,302	90.1 (± 1.2)	17.2 (± 1.2)	88.5 (± 1.9)	20.1 (± 1.8)
Age group					
18-34 years (ref)	2,782	30.2 (± 1.6)	20.6 (± 2.8)	27.7 (± 2.4)	21.3 (± 4.3)
35-64 years	4,630	50.2 (± 1.5)	**16.0 (± 1.5)**	54.2 (± 2.3)	19.3 (± 2.3)
65+ years	1,807	19.6 (± 0.9)	**15.6 (± 1.7)**	18.2 (± 1.4)	20.0 (± 2.9)

Notes:
For demographics, individual rows are calculated as a percentage of the total for the row only, not as a percentage of the column header.
Bold = Significance at the 0.05-level.
POC = People of color.

Awareness of Problem Gambling Resources

To understand the extent to which individuals are aware of Washington State's problem gambling resources, the survey asked respondents to select all resources that the individual was familiar with. Resources included support groups, clinical services, and helplines in Washington State.

To gauge the validity of responses, the survey also included a fictitious program, Gamblers Come Together (WA State).

Exhibit 11 provides the number of respondents in the full sample and the number aware of each resource. For those respondents who said they were aware of each resource, we report the percentage of the full population as well as those who were gamblers and problem gamblers.

For example, 24.3% of all respondents were aware of Gamblers Anonymous. Of those who had heard of Gamblers Anonymous, 53.5% were gamblers and 2.7% were problem gamblers.

Exhibit 11 also highlights the two most recognized resources. In addition to Gamblers Anonymous,15.8% said they were aware of the Problem Gambling Helpline. Comparatively, awareness of other resources varied from 1.1%-6.1%.

With the exception of Community Behavioral Health Programs, gamblers made up most of the respondents who were aware of problem gambling resources. WSIPP was precluded from analyzing awareness of resources by demographics due to small sample sizes.

Exhibit 11

Awareness of Resources

	Weighted N	Full population (N=9,215)	Percent aware of resources	
			Gamblers (N=4,101)	Problem gamblers (N=130)
Full sample	9,215	100	44.5 (± 1.5)	1.4 (± 0.4)
Support programs				
Gamblers Anonymous	2,238	**24.3 (± 1.2)**	53.5 (± 2.8)	2.7 (± 1.1)
Gam-Anon	2,21	2.4 (± 0.5)	59.1 (± 9.3)	6.1 (± 4.7)
Resources for clinical treatment and other services				
Evergreen Council on Problem Gambling	103	1.1 (± 0.3)	71.6 (± 11.8)	N too small
Problem Gambling Helpline	1,456	**15.8 (± 1.1)**	60.8 (± 3.5)	25.0 (± 11.4)
Community Behavioral Health Programs	559	6.1 (± 0.7)	45.4 (± 5.9)	17.9 (± 12.6)
Tribal Behavioral Health Programs	408	4.4 (± 0.6)	55.6 (± 6.6)	N too small
WA State Problem Gambling Program	441	4.8 (± 0.6)	67.8 (± 6.2)	13.3 (± 9.9)
Fictitious program				
Gamblers Come Together (WA State)	102	1.1 (± 0.3)	53.5 (± 15.4)	N too small

Notes:
For this table, we did not conduct tests of significance.
Bold Indicates the two most commonly recognized resources.

Study Question 5—What are the beliefs and attitudes about gambling?

We analyzed respondents' beliefs about the benefits and harms of gambling to identify the general beliefs about gambling at the state level and to determine if there were differences in beliefs by gambling status, region, age, gender, race, and ethnicity.

In Exhibit 12, we include the number of respondents in the sample, the total number who endorsed each belief option, and the number of respondents in each demographic group. The proportion of the population within demographic groups and their relative beliefs about gambling are also provided. For example, 5.2% of non-gamblers believed that the benefits of gambling outweighed the harms, which is significantly fewer than the 10.5% of all gamblers who believed the same.

Beliefs about gambling significantly differed by region and gender. Respondents in the North Central and Greater Columbia regions were more likely than those from King County (the reference group) to believe the benefits of gambling outweighed the harms; whereas those from the Salish and Thurston-Mason regions were significantly less likely than those from King County to believe the same. Men were significantly more likely than women to believe the benefits of gambling outweighed the harms. In contrast, women were significantly more likely than men to believe that the harms somewhat or far outweighed the benefits.

Exhibit 12

Beliefs About Gambling, by Gambling Status and Demographic Group

	Weighted N	Percent of population (N=8,895)	Percent (±) margin of error: Harms somewhat or far outweigh the benefits (N=6,031)	Benefits somewhat or far outweigh the harms (N=670)	The benefits and harms are about equal (N=2,195)
Percent of population	8,895	100	67.8 (± 1.5)	7.5 (± 0.9)	24.7 (± 1.3)
Gambling status					
Non-gamblers (ref)	4,932	55.4 (± 1.5)	77.3 (± 1.7)	5.2 (± 1.0)	17.5 (± 1.5)
All gamblers	3,964	44.6 (± 1.5)	**56.0 (± 2.3)**	**10.5 (± 1.5)**	**33.6 (± 2.2)**
Problem gamblers	126	1.4 (± 0.4)	64.6 (± 14.3)	N too small	28.8 (± 13.3)
Age					
18-34 years (ref)	2,756	31.0 (± 1.7)	67.6 (± 3.3)	7.7 (± 1.9)	24.7 (± 3.0)
35-64 years	4,421	49.7 (± 1.5)	66.7 (± 1.9)	7.6 (± 1.1)	25.7 (± 1.7)
65+ years	1,718	19.3 (± 0.9)	71.0 (± 2.2)	7.0 (± 0.9)	22.0 (± 1.3)
Gender					
Men (ref)	4,426	49.8 (± 1.5)	65.1 (± 2.2)	9.2 (± 1.4)	25.7 (± 2.0)
Women	4,469	50.2 (± 1.5)	**70.4 (± 1.9)**	**5.9 (± 1.1)**	23.7 (± 1.7)
Race					
White (ref)	6,928	77.9 (± 1.4)	67.6 (± 1.6)	7.3 (± 0.9)	25.1 (± 1.5)
POC	1,967	22.1 (± 1.4)	68.7 (± 3.5)	8.3 (± 2.1)	23.0 (± 3.2)
Ethnicity					
Hispanic (ref)	877	9.9 (± 1.2)	66.0 (± 6.1)	9.5 (± 4.0)	24.5 (± 5.4)
Non-Hispanic	8,018	90.1 (± 1.2)	68.0 (± 1.5)	7.3 (± 0.8)	24.7 (± 1.4)
MCO Region					
King (ref)	2,704	30.4 (± 1.0)	69.0 (± 2.5)	7.7 (± 1.5)	26.3 (± 2.3)
Great Rivers	346	3.9 (± 0.5)	60.0 (± 8.2)	11.6 (± 6.1)	28.4 (± 7.3)
Greater Columbia	803	9.0 (± 0.7)	62.3 (± 5.4)	**11.4 (± 1.5)**	26.3 (± 4.8)
North Central	289	3.3 (± 0.5)	58.0 (± 10.0)	**17.5 (± 8.2)**	24.5 (± 8.6)
North Sound	1,523	17.1 (± 0.9)	70.0 (± 3.4)	6.2 (± 1.8)	23.8 (± 3.1)
Pierce	1,037	11.7 (± 0.8)	70.9 (± 4.5)	6.4 (± 2.4)	22.7 (± 4.1)
Salish	466	5.2 (± 0.5)	72.9 (± 5.5)	**5.0 (± 1.9)**	22.2 (± 5.3)
Southwest	589	6.6 (± 0.5)	66.4 (± 5.5)	5.7 (± 2.6)	27.9 (± 5.3)
Spokane	714	8.0 (± 0.6)	65.1 (± 4.9)	5.8 (± 2.4)	29.0 (± 4.6)
Thurston-Mason	425	4.8 (± 0.5)	69.3 (± 6.4)	**4.5 (± 2.4)**	26.2 (± 6.2)

Note:
Bold = Significance at the 0.05-level.
POC = People of color.
MCO = Managed Care Organization.

Opinions about the availability of gambling were also analyzed at the state level and across demographic groups. Exhibit 13 lays out the number of respondents in the sample, the total number who endorsed each opinion, and the number of respondents in each demographic group. Also provided is the proportion of the population within demographic groups and their relative opinions about gambling.

Opinions about gambling availability significantly differed by gambling status and age. Gamblers and problem gamblers were significantly more likely than non-gamblers to think gambling was not available enough. Each age group differed significantly from one another. Respondents 18 to 34 years old were less likely to think gambling was too widely available and more likely to believe the current availability of gambling was fine. Respondents 65 years and older were most likely to think gambling was too widely available and least likely to think gambling availability was fine or not available enough.

Exhibit 13

Opinions About Gambling Availability

	Weighted N	Percent of population	Percent (±) margin of error: Gambling is too widely available	Gambling is not available enough	Current availability of gambling is fine
Percent of population	8,890	100	31.2 (± 1.4)	4.6 (± 0.7)	64.2 (± 1.5)
Gambling status					
Non-gamblers (ref)	4,888	55.0 (± 1.5)	39.3 (± 2.0)	2.3 (± 0.8)	58.5 (± 2.1)
All gamblers	4,003	44.6 (± 1.5)	**21.5 (± 1.9)**	**7.5 (± 1.3)**	**71.1 (± 2.2)**
Problem gamblers	124	1.4 (± 0.4)	31.3 (± 12.7)	**16.9 (± 12.9)**	51.8 (± 14.8)
Gender					
Men (ref)	4,474	49.7 (± 1.5)	29.5 (± 2.1)	6.9 (± 1.3)	63.6 (± 2.2)
Women	4,469	50.3 (± 1.5)	**33.0 (± 2.0)**	**2.3 (± 0.8)**	64.7 (± 2.0)
Race					
White (ref)	6,931	78.0 (± 1.4)	31.7 (± 1.6)	4.2 (± 0.8)	64.0 (± 1.6)
POC	1,960	22.0 (± 1.4)	29.4 (± 3.4)	6.0 (± 2.0)	64.6 (± 3.6)
Ethnicity					
Hispanic (ref)	880	9.9 (± 1.2)	33.2 (± 6.2)	6.6 (± 3.5)	60.2 (± 6.4)
Non-Hispanic	8,010	90.1 (± 1.2)	31.0 (± 1.4)	4.4 (± 0.7)	64.6 (± 1.5)
Age					
18-34 years (ref)	2,731	30.7 (± 1.6)	*24.5 (± 3.2)†*	5.8 (± 1.7)	*69.7 (± 3.3)†*
35-64 years	4,457	50.1 (± 1.5)	*31.0 (± 1.9)†*	4.7 (± 1.0)	*64.3 (± 2.0)†*
65+ years	1,703	19.2 (± 0.9)	*42.6 (± 2.4)†*	**2.4 (± 0.9)**	*55.0 (± 2.4)†*
MCO region					
King (ref)	2,689	30.3 (± 1.0)	26.8 (± 2.4)	4.7 (± 1.3)	64.7 (± 2.5)
Great Rivers	346	3.9 (± 0.5)	**36.0 (± 8.2)**	6.7 (± 5.2)	57.3 (± 8.5)
Greater Columbia	805	9.1 (± 0.7)	27.9 (± 4.7)	7.5 (± 3.2)	64.7 (± 5.2)
North Central	285	3.2 (± 0.5)	24.5 (± 7.9)	7.4 (± 5.3)	68.1 (± 8.8)
North Sound	1,523	17.1 (± 0.9)	**33.9 (± 3.5)**	3.1 (± 1.4)	63.0 (± 3.6)
Pierce	1,038	11.7 (± 0.8)	**38.4 (± 4.7)**	5.6 (± 2.9)	**56.0 (± 4.9)**
Salish	464	5.2 (± 0.5)	**36.5 (± 6.3)**	3.5 (± 2.1)	60.1 (± 6.4)
Southwest	608	6.8 (± 0.6)	28.5 (± 5.8)	4.2 (± 2.7)	67.3 (± 5.9)
Spokane	709	8.0 (± 0.6)	31.8 (± 4.8)	2.9 (± 2.1)	65.3 (± 5.0)
Thurston-Mason	423	4.8 (± 0.5)	**36.4 (± 6.7)**	3.4 (± 2.2)	60.3 (± 6.8)

Notes:
† indicates that all demographic groups are significantly different from each other.
Bold = Significance at the 0.05-level.
MCO = Managed Care Organization.
POC = People of color.

III. Summary and Limitations

We found that in 2021, 44.7% of Washington State adults gambled in the previous 12 months. Among those who gambled, 3.5% were problem gamblers (1.4% of the total adult population).

Rates of gambling varied by gender, marital status, ethnicity, age, education, military service, employment, type of insurance, and geographic region.

We observed that of those who gambled, online gamblers constituted 27.8% (online gamblers may also gamble in brick-and-mortar facilities). We also determined that the rate of problem gambling is significantly higher for online gamblers than for those who gamble only in brick-and-mortar establishments (7.8% vs 1.7%).

Due to the small prevalence of problem gambling in the overall adult population (1.4%), a limitation of the survey was that we were unable to detect a large number of statistically significant differences in the prevalence rate for problem gambling (based on the ten demographics collected). This was also an issue for establishing if people seeking problem gambling treatment were able to find it. A future focused panel study, or larger survey population study, might glean more specific information about problem gamblers based on demographics.

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